teristic function. The p-ordered characteristic function is defined to be

$$\chi_{\rho,p}(\alpha) = \mathrm{Tr}[\hat{\rho}\hat{D}(\alpha)]e^{\frac{1}{2}p|\alpha|^2} = \mathrm{Tr}[\hat{\rho}e^{\alpha\hat{a}^\dagger - \alpha^*\hat{a}}]e^{\frac{1}{2}p|\alpha|^2}, \tag{3.90}$$

where the parameter p can take three values, $1, 0$, and -1. Each value corresponds to a particular ordering of creation and annihilation parameters in the expression. Specifically, it can be shown that

$$\chi_{\rho,1}(\alpha) = \mathrm{Tr}[\hat{\rho}\, e^{\alpha\hat{a}^\dagger} e^{-\alpha^*\hat{a}}] \qquad \text{(normal)}, \tag{3.91}$$

$$\chi_{\rho,0}(\alpha) = \mathrm{Tr}[\hat{\rho}\, e^{\alpha\hat{a}^\dagger - \alpha^*\hat{a}}] \qquad \text{(symmetric)}, \tag{3.92}$$

$$\chi_{\rho,-1}(\alpha) = \mathrm{Tr}[\hat{\rho}\, e^{-\alpha^*\hat{a}} e^{\alpha\hat{a}^\dagger}] \qquad \text{(anti-normal)}, \tag{3.93}$$

such that $p = 1$ corresponds to normal order ($\hat{a}^\dagger$ before $\hat{a}$), $p = -1$ to anti-normal order ($\hat{a}$ before $\hat{a}^\dagger$), and $p = 0$ to symmetric order (average of all possible orders). The quasiprobability distributions are given by a two-dimensional Fourier transform of the characteristic function

$$\frac{1}{\pi^2}\int \chi_{\rho,p}(\beta)e^{\alpha\beta^* - \alpha^*\beta}d^2\beta = \begin{cases} P_\rho(\alpha) & \text{for } p = 1 \\ W_\rho(\alpha) & \text{for } p = 0 \\ Q_\rho(\alpha) & \text{for } p = -1 \end{cases}, \tag{3.94}$$

where the integral is over the complex plane and $W(\alpha)$ should be interpreted as $W(x, y)$ with $\alpha = (x + iy)/\sqrt{2}$.

Characteristic functions and quasiprobability distributions are powerful tools with many uses in theoretical quantum optics and beyond. This is outside the scope of these notes, however, and we will not delve further into the details. For more information see e.g. the books by Barnett and Radmore [1] or Gerry and Knight [5]. Here, we will only be needing the Wigner function.

3.1.8 Squeezed states

So far, we have discussed Fock states and coherent states. There are of course many other states of the harmonic oscillator which are interesting and useful for quantum technology. Among these, the so-called *squeezed states* are of particular importance, because they are relatively easy to generate and at the same time enable superclassical performance in many applications.

In general, squeezing can be defined in any system (not just harmonic oscillators). Given two observables $\hat{A}$ and $\hat{B}$, the uncertainty relation (2.24) states that $\langle\Delta^2\hat{A}\rangle\langle\Delta^2\hat{B}\rangle \geq \frac{1}{4}|\langle[\hat{A}, \hat{B}]\rangle|^2$ in any state. A state is said to be squeezed if

$$\langle\Delta^2\hat{A}\rangle < \frac{1}{2}|\langle[\hat{A}, \hat{B}]\rangle| \qquad \text{or} \qquad \langle\Delta^2\hat{B}\rangle < \frac{1}{2}|\langle[\hat{A}, \hat{B}]\rangle|, \tag{3.95}$$

i.e. one of the observables is contributing less uncertainty than an equal share of the minimum joint uncertainty. When the system consists of harmonic oscillators (one or several) and $\hat{A}$ and $\hat{B}$ are chosen to be (combinations of) quadratures, this is referred to as *quadrature squeezing*. This is the only kind of squeezing we will be concerned with here.

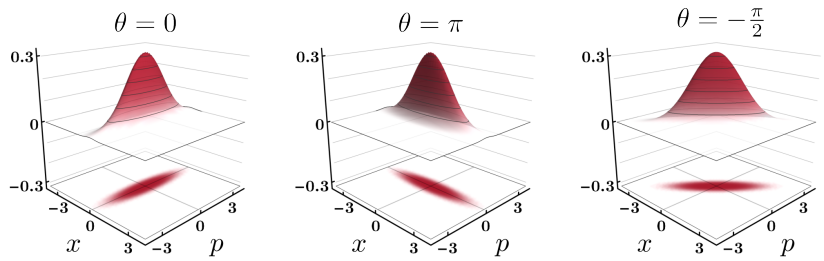

Figure 3.5: Wigner functions for single-mode squeezed states with $r = 0.7$ (about 6 dB) and squeezing in $\hat{x}$ (left), $\hat{p}$ (middle), and $(\hat{x} - \hat{p})/\sqrt{2}$ (right).

Single-mode squeezed vacuum

Consider first a single harmonic oscillator and take the observables to be $\hat{x}$ and $\hat{p}$. Since $[\hat{x}, \hat{p}] = i$, there will be quadrature squeezing whenever $\langle \Delta^2 \hat{x} \rangle$ or $\langle \Delta^2 \hat{p} \rangle$ is less than $\frac{1}{2}$, the variance in a coherent state.

The *single-mode squeezing operator* is defined as

$$\hat{S}(\xi) = e^{\frac{1}{2}(\xi^* \hat{a}^2 - \xi (\hat{a}^\dagger)^2)}, \tag{3.96}$$

where $\xi = re^{i\theta}$ is any complex number. This operator is unitary with $\hat{S}^\dagger(\xi) = \hat{S}(-\xi)$. The annihilation operator transforms as

$$\hat{S}^\dagger(\xi) \hat{a} \hat{S}(\xi) = \cosh(r)\hat{a} - e^{i\theta} \sinh(r) \hat{a}^\dagger. \tag{3.97}$$

For the quadratures therefore

$$\hat{S}(\xi)^\dagger \hat{x} \hat{S}(\xi) = [c_r - \cos(\theta) s_r]\hat{x} - \sin(\theta) s_r \hat{p}, \tag{3.98}$$

$$\hat{S}(\xi)^\dagger \hat{p} \hat{S}(\xi) = [c_r + \cos(\theta) s_r]\hat{p} - \sin(\theta) s_r \hat{x}, \tag{3.99}$$

where $c_r = \cosh(r)$ and $s_r = \sinh(r)$. For the particular case $\theta = 0$, the squeezing operation takes $\hat{x} \to e^{-r}\hat{x}$ and $\hat{p} \to e^r \hat{p}$ and therefore

$$\langle \Delta^2 \hat{x} \rangle \to e^{-2r} \langle \Delta^2 \hat{x} \rangle, \tag{3.100}$$

$$\langle \Delta^2 \hat{p} \rangle \to e^{2r} \langle \Delta^2 \hat{p} \rangle. \tag{3.101}$$

That is, the squeezing reduces the uncertainty in $\hat{x}$ by a factor of e^{-2r} and increases the uncertainty in $\hat{p}$ by the inverse of the same factor. We say that it 'squeezes' $\hat{x}$ and 'anti-squeezes' $\hat{p}$. In general, $\hat{S}(\xi)$ will squeeze the quadrature $\hat{x}(\frac{\theta}{2})$ and anti-squeeze $\hat{p}(\frac{\theta}{2})$

$$\langle \Delta^2 \hat{x}(\theta/2) \rangle \to e^{-2r} \langle \Delta^2 \hat{x}(\theta/2) \rangle, \tag{3.102}$$

$$\langle \Delta^2 \hat{p}(\theta/2) \rangle \to e^{2r} \langle \Delta^2 \hat{p}(\theta/2) \rangle. \tag{3.103}$$

The squeezing factor $s = e^{-2r}$ quantifies the amount of squeezing. In experimental works, it is often reported in *decibels* (dB), defined as $10 \log_{10}(s)$. Squeezing by a factor of 2 corresponds to about 3 dB.

Squeezing can in principle be applied to any state. However, as we will see, the squeezed states that are most readily generated correspond to applying

$\hat{S}(\xi)$ to the vacuum. These states $|\xi\rangle = \hat{S}(\xi)|0\rangle$ are known as *single-mode squeezed vacuum states*. What do these states look like in phase space? Since the expectation values of the quadratures vanish in the vacuum state, they are also zero for the squeezed vacuum, while the variances change. The symmetric gaussian distribution of the vacuum gets squeezed along the direction of $\hat{x}(\frac{\theta}{2})$ and anti-squeezed along $\hat{p}(\frac{\theta}{2})$. The Wigner function is

$$W_{|\xi\rangle}(x,p) = \frac{1}{\pi} e^{-e^{2r}[x\cos\left(\frac{\theta}{2}\right)+p\sin\left(\frac{\theta}{2}\right)]^2 - e^{-2r}[p\cos\left(\frac{\theta}{2}\right)-x\sin\left(\frac{\theta}{2}\right)]^2}. \tag{3.104}$$

Some Wigner functions of single-mode squeezed states are plotted in Fig. 3.5.

How about the expansion in the Fock basis? Considering the Taylor expansion of the squeezing operator (3.96), it is clear that all terms contain an even number of annihilation and creation operators. Applying squeezing to the vacuum will therefore generate a state containing only even numbers of excitations. Specifically, in the Fock basis the state is

$$|\xi\rangle = \hat{S}(\xi)|0\rangle = \frac{1}{\sqrt{\cosh(r)}} \sum_{k=0}^{\infty} \frac{\sqrt{(2k)!}}{2^k k!} e^{ik\theta} \tanh^k(r)|2k\rangle. \tag{3.105}$$

The mean excitation number depends on the strength of the squeezing but not on the angle θ. Using $\hat{n} = \frac{1}{2}(\hat{x}^2 + \hat{p}^2 - 1)$ and that the vacuum variances are $\frac{1}{2}$, it can be seen that

$$\langle\hat{n}\rangle = \langle\xi|\hat{n}|\xi\rangle = \frac{1}{2}\left(\frac{1}{2}e^{-2r} + \frac{1}{2}e^{2r} - 1\right) = \sinh^2(2r). \tag{3.106}$$

As expected, $\langle\hat{n}\rangle = 0$ when there is no squeezing ($r = 0$). On the other hand, as the squeezing grows $r \to 0$, the excitation number and hence also the mean energy $\langle\hat{H}\rangle$ diverges. Infinitely squeezed states thus require infinite energy and are therefore not physical – they are impossible to realise exactly.

Two-mode squeezed vacuum

It is also possible to have quadrature squeezing across several oscillators. This correlates the quadratures of one system with those of another and generates entanglement. Consider a pair of harmonic oscillators and the joint observables $\hat{x}_\pm = (\hat{x}_A \pm \hat{x}_B)/\sqrt{2}$ and $\hat{p}_\pm = (\hat{p}_A \pm \hat{p}_B)/\sqrt{2}$, where subscripts A and B label the two oscillators. These operators fulfill the commutation relations

$$[\hat{x}_+, \hat{p}_+] = [\hat{x}_-, \hat{p}_-] = i, \qquad [\hat{x}_+, \hat{p}_-] = [\hat{x}_-, \hat{p}_+] = 0, \tag{3.107}$$

and of course $[\hat{x}_+, \hat{x}_-] = [\hat{p}_+, \hat{p}_-] = 0$. Thus $\hat{x}_+$ and $\hat{p}_+$ are conjugate variables, as are $\hat{x}_-$ and $\hat{p}_-$, but $\hat{x}_+$ and $\hat{p}_-$ or $\hat{x}_-$ and $\hat{p}_+$ can be simultaneously measured. It is therefore possible, for example, to squeeze $\hat{x}_+$ and $\hat{p}_-$ at the same time, while anti-squeezing $\hat{p}_+$ and $\hat{x}_-$. This is exactly the type of effect two-mode squeezing has.

The *two-mode squeezing operator* is defined as

$$\hat{S}_2(\xi) = e^{\xi^*\hat{a}\hat{b} - \xi\hat{a}^\dagger\hat{b}^\dagger} \tag{3.108}$$

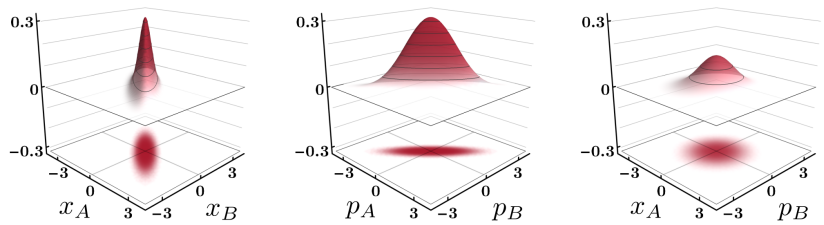

Figure 3.6: Marginals of the Wigner functions for a two-mode squeezed vacuum state with $r = 0.7$ (about 6 dB) and $\theta = 0$. Note that $\hat{x}_A$, $\hat{x}_B$ are anti-correlated, $\hat{p}_A$, $\hat{p}_B$ are correlated, and $\hat{x}_A$, $\hat{p}_B$ have no correlation.

where $\xi = re^{i\theta}$ is any complex number and $\hat{a}$ and $\hat{b}$ are the annihilation operators on systems A and B, respectively. Under two-mode squeezing, they transform as

$$\hat{S}_2^\dagger(\xi)\hat{a}\hat{S}_2(\xi) = \cosh(r)\hat{a} - e^{i\theta}\sinh(r)\hat{b}^\dagger, \tag{3.109}$$

$$\hat{S}_2^\dagger(\xi)\hat{b}\hat{S}_2(\xi) = \cosh(r)\hat{b} - e^{i\theta}\sinh(r)\hat{a}^\dagger, \tag{3.110}$$

and for the quadratures

$$\hat{S}_2^\dagger(\xi)\hat{x}_+\hat{S}_2(\xi) = [c_r - \cos(\theta)s_r]\hat{x}_+ - \sin(\theta)s_r\hat{p}_+, \tag{3.111}$$

$$\hat{S}_2^\dagger(\xi)\hat{x}_-\hat{S}_2(\xi) = [c_r + \cos(\theta)s_r]\hat{x}_- + \sin(\theta)s_r\hat{p}_-, \tag{3.112}$$

$$\hat{S}_2^\dagger(\xi)\hat{p}_+\hat{S}_2(\xi) = [c_r + \cos(\theta)s_r]\hat{p}_+ - \sin(\theta)s_r\hat{x}_+, \tag{3.113}$$

$$\hat{S}_2^\dagger(\xi)\hat{p}_-\hat{S}_2(\xi) = [c_r - \cos(\theta)s_r]\hat{p}_+ + \sin(\theta)s_r\hat{x}_+, \tag{3.114}$$

where again $c_r = \cosh(r)$ and $s_r = \sinh(r)$. For $\theta = 0$, we have $\hat{x}_+ \to e^{-2r}\hat{x}_+$ and $\hat{p}_- \to e^{-2r}\hat{p}_-$ while $\hat{x}_- \to e^{2r}\hat{x}_-$ and $\hat{p}_+ \to e^{2r}\hat{p}_+$, so

$$\langle\Delta^2\hat{x}_+\rangle \to e^{-2r}\langle\Delta^2\hat{x}_+\rangle, \tag{3.115}$$

$$\langle\Delta^2\hat{x}_-\rangle \to e^{2r}\langle\Delta^2\hat{x}_-\rangle, \tag{3.116}$$

$$\langle\Delta^2\hat{p}_+\rangle \to e^{2r}\langle\Delta^2\hat{p}_+\rangle, \tag{3.117}$$

$$\langle\Delta^2\hat{p}_-\rangle \to e^{-2r}\langle\Delta^2\hat{p}_-\rangle. \tag{3.118}$$

That is, $\hat{x}_+$ and $\hat{p}_-$ are being squeezed, while $\hat{x}_-$ and $\hat{p}_+$ are anti-squeezed. In general, $\hat{S}_2(\xi)$ will squeeze $\hat{x}_+(\theta/2)$ and $\hat{p}_-(\theta/2)$ and anti-squeeze $\hat{x}_-(\theta/2)$ and $\hat{p}_+(\theta/2)$, where $\hat{x}_+(\theta/2) = (\hat{x}_A(\theta/2) + \hat{x}_B(\theta/2))/\sqrt{2}$ etc.

When applied to the joint vacuum state of the two oscillators, $\hat{S}_2(\xi)$ generates a *two-mode squeezed vacuum state* $|\xi\rangle_2 = \hat{S}_2(\xi)|0,0\rangle_{AB}$. Like for the single-mode squeezed vacuum, the expectation values of all quadratures are zero in the state $|\xi\rangle_2$, while the variances differ from those of the vacuum. The Wigner function of the two-mode squeezed vacuum can be expressed as a product of singe-mode squeezed vacuum Wigner functions in the normal modes. E.g. for $\theta = 0$ we have

$$W_{|\xi\rangle_2}(x_A, p_A, x_B, p_B) = W_{|\xi\rangle_1}(x_+, p_+)W_{|\xi\rangle_1}(p_-, -x_-), \tag{3.119}$$

where $x_\pm = (x_A \pm x_B)/\sqrt{2}$ and $p_\pm = (p_A \pm p_B)/\sqrt{2}$. Since this is now a function of four variables, we cannot plot it directly. However, we can plot

its marginals, obtained by integrating out two variables, e.g. $W_{|\xi\rangle_2}(x_A, x_B) = \int W_{|\xi\rangle_2}(x_A, p_A, x_B, p_B) dp_A dp_B$. Plots of some marginals are shown in Fig. 3.6.

It is clear that two-mode squeezed states display correlations between the quadratures of modes A and B. Specifically, $\hat{p}_A(\theta/2)$ and $\hat{p}_B(\theta/2)$ will be correlated while $\hat{x}_A(\theta/2)$ and $\hat{x}_B(\theta/2)$ will be anti-correlated, as can be seen in Fig. 3.6 for $\theta = 0$. In the limit of infinite squeezing ($r \to \infty$), the correlations become perfect, i.e. $\hat{p}_A(\theta/2) = \hat{p}_B(\theta/2)$ and $\hat{x}_A(\theta/2) = -\hat{x}_B(\theta/2)$. These correlations are quantum in nature and in fact the two-mode squeezed states are entangled.

In the Fock basis, the two-mode squeezed vacuum state exhibit perfect correlation between the excitation numbers in the two modes. It can be expressed

$$|\xi\rangle_2 = \operatorname{sech}(r) \sum_{n=0}^{\infty} (-1)^n e^{i\theta n} \tanh^n(r) |n, n\rangle_{AB}. \tag{3.120}$$

We see that the excitation numbers in modes A and B are equal in all terms. For low squeezing ($r \ll 1$),

$$|\xi\rangle_2 \approx \frac{1}{\sqrt{1+r^2}} \left(|0,0\rangle - e^{i\theta} r |1,1\rangle \right), \tag{3.121}$$

where we have dropped terms of order $O(r^2)$ or higher. In this limit, $|\xi\rangle_2$ can thus be understood as a partially entangled state of two qubits. In the opposite limit of infinite squeezing ($r \to \infty$), $\tanh(r)$ tends to 1 and thus $|\xi\rangle_2$ tends towards an equally weighted superposition of all excitation numbers. Such a superposition is not normalisable – the limit of infinite squeezing is not physical since it would require infinite energy. However, considering maximally entangled states in dimension d, such as

$$\frac{1}{\sqrt{d}} \left(|0,0\rangle + |1,1\rangle + \cdots + |d-1, d-1\rangle \right), \tag{3.122}$$

the limit of infinite squeezing for $|\xi\rangle_2$ can be understood as an infinite-dimensional maximally entangled state.

3.1.9 Cat states

Coherent states, single- and to-mode squeezed states, as well as the thermal states discussed in the next section all belong to the set of Gaussian states. ...

An interesting example of non-Gaussian states are superpositions of coherent states. Coherent-state superpositions are often known as *cat states* in analogy with the famous thought experiment of Schrödinger because, for large amplitudes, different coherent states are macroscopically distinct (and coherent states are also in a sense the 'most classical' quantum states of light, as discussed in Sec. 3.1.5). In particular, we can define the so-called even and odd cat states

$$|C_{\pm}^{\alpha}\rangle = \frac{1}{N_{\pm}} (|\alpha\rangle \pm |-\alpha\rangle), \tag{3.123}$$